Hamilton School for
Classical and Civic Education
UNIVERSITY of FLORIDA

Calculation, Knowledge, and the Entrepreneur

Capitalism and Its Critics
ISS 2440 — Intro to PPEL

Louis Rouanet

Fall 2026

Armen Alchian and the H-bomb

Armen Alchian, interview, quoted in Newhard (2014)

"We knew they were developing this H-bomb, but we wanted to know, what's in it? What's the fissile material? Well there's thorium, thallium, beryllium, and something else, and we asked Herman Kahn and he said, 'Can't tell you'... I said, 'I'll find out.'"

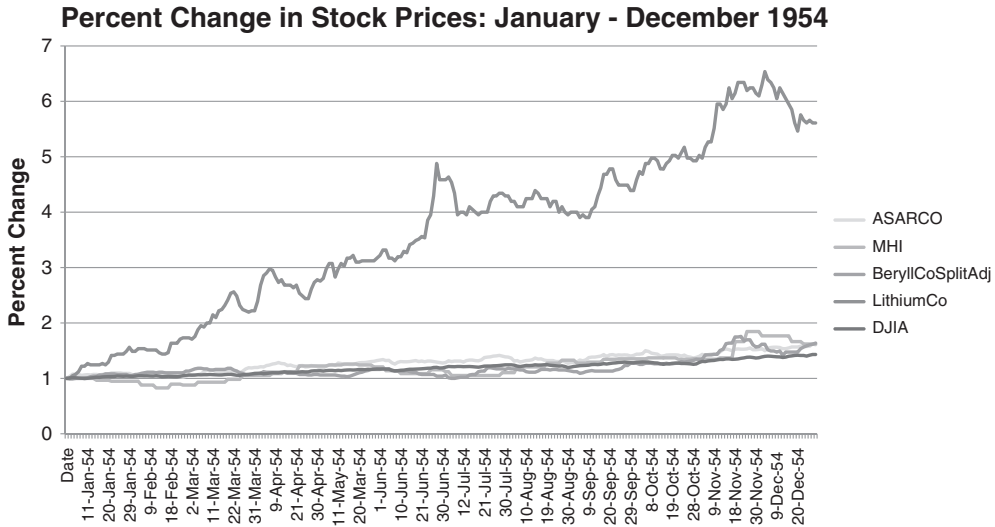
Castle Bravo, March 1, 1954

Armen Alchian and the H-bomb

Armen Alchian, interview, quoted in Newhard (2014)

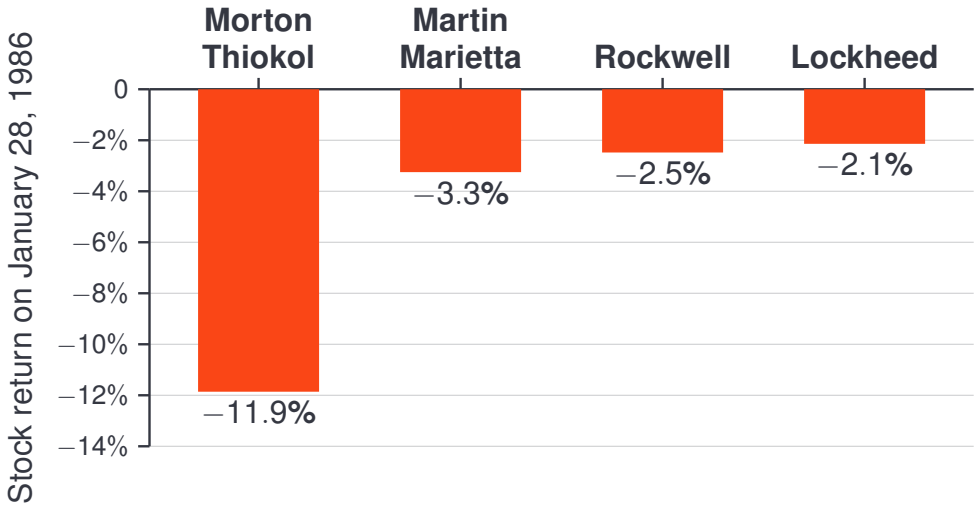
"Look up for me the price of these companies... and here were these four or five stocks going like this, and then about, I think it was September, ... one of them started to go like that, from $2 to around $10, the rest were going like this, so I thought 'Well, that's interesting'... I wrote it up and distributed it around the social science group the next day. I got a phone call from the head of RAND calling me in... he said 'Armen, we've got to suppress this'... I said 'Yes, sir', and I took it and put it away, and that was the first event study."

Armen Alchian
(1914–2013)



Note: Stock-price indices, January 1954 = 1 (despite the original chart's "Percent Change" label). Newhard (2014), Chart 7, reproduced from the vector original.

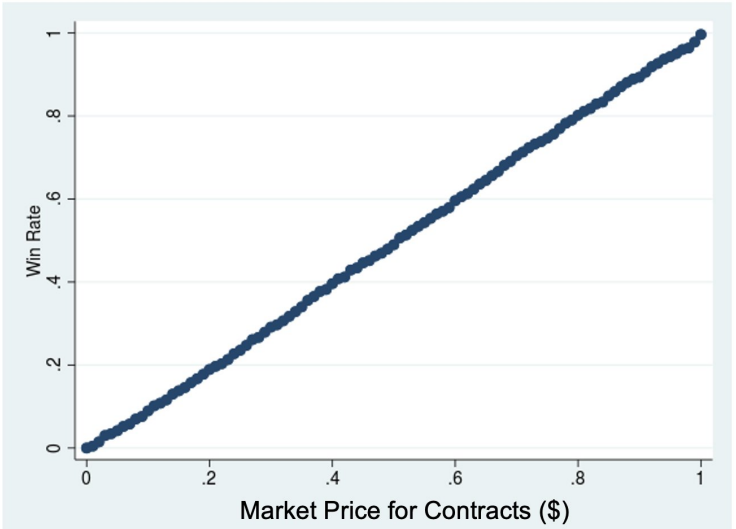
Challenger, January 28, 1986



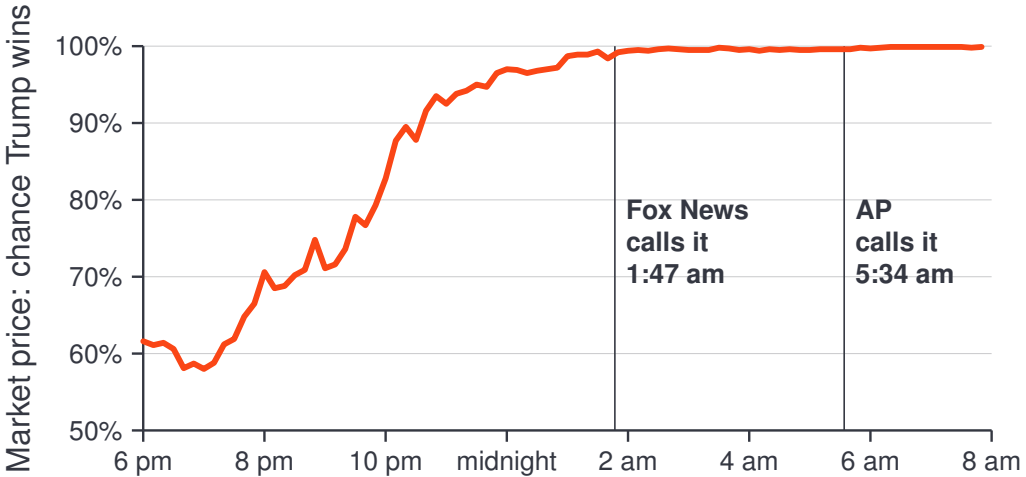
Note: Stock returns on January 28, 1986, from the previous close to the close of trading, for the four main shuttle contractors.
Maloney, Michael T. and J. Harold Mulherin. (2003). “The Complexity of Price Discovery in an Efficient Market: The Stock Market Reaction to the Challenger Crash,” *Journal of Corporate Finance* 9(4), 453–479.

Who knew?

- Thirteen minutes after the explosion, trading in Morton Thiokol was halted on the New York Stock Exchange. No one had said a word about the cause.
- Morton Thiokol made the solid rocket boosters. Five months later the Rogers Commission blamed their O-rings, which failed in the cold.
- Maloney and Mulherin found no evidence of insider trading. $\Rightarrow$ The price pointed to the responsible contractor months before the official report.

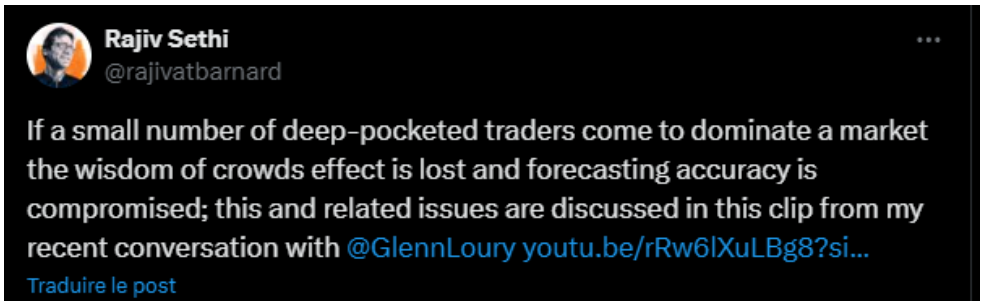


Note: Sports-betting contract prices against realized win rates.



Note: Price of “Will Donald Trump win the 2024 US Presidential Election?” on Polymarket, election night, November 5–6, 2024, Eastern time, every 10 minutes. Polymarket price history (CLOB API).

The French whale: the criticism



The French whale: the criticism

The French whale

- In 2024 a French trader, “Théo”, bet more than $30 million on Trump on Polymarket.
- He said he paid for his own polls, asking people whom their neighbors would vote for, not whom they would vote for.
- He made about $80 million. $\Rightarrow$ His research informed his bets, which moved the price for everyone else.

Osipovich (2024), *The Wall Street Journal*; Chainalysis (2024).

Manipulation and market accuracy

ELSEVIER

Journal of Economic Behavior & Organization

Volume 60, Issue 4, August 2006, Pages 449-459

Information aggregation and manipulation in an experimental market

Robin Hanson, Ryan Oprea, David Porter

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Abstract

Prediction markets are increasingly being considered as methods for gathering, summarizing and aggregating diffuse information by governments and businesses alike. Critics worry that these markets are susceptible to price manipulation by agents who wish to distort decision making. We study the effect of manipulators on an experimental market, and find that manipulators are unable to distort price accuracy. Subjects without manipulation incentives compensate for the bias in offers from manipulators by setting a different threshold at which they are willing to accept trades.

- In the experiment, attempts to distort prices did not reduce accuracy.
- A distorted price creates a profit opportunity for traders with better information.
- Hanson and Oprea (2009): this can encourage information gathering and improve accuracy.

How can a price know
what no one knows?

What is socialism?

- Socialism: collective ownership of the means of production.
- Means of production: land, factories, machines and other assets used to produce goods and services.
- In the calculation debate, these assets are controlled collectively rather than bought and sold by private owners.

Incentives vs. knowledge

- Incentives: who takes the trash out? What makes people do the work and use resources carefully?

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- Knowledge: even if everyone wants to help, how do we know what to produce, how much, and by which method?

Incentives vs. knowledge

- Incentives: who takes the trash out? What makes people do the work and use resources carefully?
- Knowledge: even if everyone wants to help, how do we know what to produce, how much, and by which method?
- The calculation debate asks whether planners can make these choices without market prices for productive assets.

Central planning

- Central planning: one authority decides what to produce, how, and for whom. Land, factories and machines are owned in common.
- In the USSR, Gosplan, the State Planning Committee, set five-year plans from 1928 to 1991.
- Targets came in physical units: tons of steel, meters of cloth, pairs of shoes. $\Rightarrow$ Factories met the target, not the need.

Mises and economic calculation

- Should we build a railway between two towns? Engineers can tell you how much steel, coal and labor it takes.
- They cannot tell you whether that steel, coal and labor would be worth more somewhere else. For that you need their money prices.
- Under socialism the state owns all the means of production. $\Rightarrow$ No one buys or sells them. $\Rightarrow$ No prices for them. $\Rightarrow$ Nothing to calculate with.

Ludwig von Mises
(1881–1973)

Mises and economic calculation

Ludwig von Mises, "Economic Calculation in the Socialist Commonwealth," 1920

"Thus in the socialist commonwealth every economic change becomes an undertaking whose success can be neither appraised in advance nor later retrospectively determined. There is only groping in the dark. Socialism is the abolition of rational economy."

Ludwig von Mises
(1881–1973)

Why technical knowledge is not enough

- Suppose the planner knew every technical trade-off: how much labor a machine can replace, in every use.
- In a world where nothing changes, that might be enough. Mises: “The static state can dispense with economic calculation.”

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- But the opportunity cost changes. Put a machine in one factory, and it is unavailable to other factories. $\Rightarrow$ The technical recipe does not tell us which use is worth most.
- Competing uses must be appraised, again and again, by people who put their own money on the answer: the market process.

Lange and market socialism

- Oskar Lange (1936–37): a Central Planning Board can play the market's role.
- The Board posts prices. Managers follow two rules: produce at the lowest cost, and produce until one more unit costs what it sells for.
- A shortage? Raise the price. A surplus? Lower it. $\Rightarrow$ Trial and error, like an auctioneer.
- Lange's joke: socialists should put up a statue of Mises for posing the problem.

Oskar Lange
(1904–1965)

Hayek and the knowledge problem

- Lange assumes the Board has the data. Hayek: the data are never given to any one mind.
- What matters is knowledge of “the particular circumstances of time and place”: the half-empty truck, the idle machine, the local shortage.
- Much of it is tacit: people act on it but cannot write it down, so they cannot send it to a planning board.

Friedrich Hayek
(1899–1992)

Hayek and the knowledge problem

F. A. Hayek, "The Use of Knowledge in Society," 1945

"Assume that somewhere in the world a new opportunity for the use of some raw material, say tin, has arisen, or that one of the sources of supply of tin has been eliminated. It does not matter for our purpose—and it is very significant that it does not matter—which of these two causes has made tin more scarce. All that the users of tin need to know is that some of the tin they used to consume is now more profitably employed elsewhere, and that in consequence they must economize tin."

Friedrich Hayek
(1899–1992)

Knowledge that markets create

- Often people do not know what they want, or what they believe, until they have to choose with their own money.
- Hayek (1968): competition is a discovery procedure. It brings out knowledge that did not exist before.
- Prices keep people honest. A survey answer costs nothing; a bet or a purchase does.
  $\Rightarrow$ Back to Alchian, Challenger and the election.

Two arguments against central planning

	Mises (1920)	**Hayek (1945)**
The problem	Calculation: comparing costs	Knowledge: who knows what
What markets supply	Prices for the means of production	A summary of scattered, tacit knowledge
An angel as planner?	Still fails	Still fails
A faster computer?	No: the trade-offs change as you act	No: the data are not in the computer

The critics reply

- Trial and error works. Lange's Board finds prices the way a market does.
  - Hayek (1940): who sets millions of prices, how fast, and what does a manager gain from getting them right?
- Firms already plan. Walmart and Amazon are planned economies larger than many countries.
  - They plan inside a price system: they buy, sell and count profit at market prices.
- Computers can do it. Linear programming in the USSR, Cybersyn in Chile, and now AI.

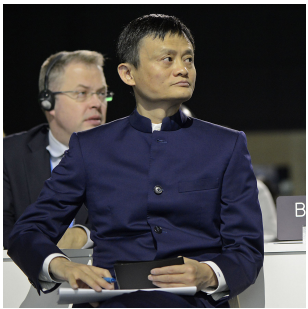
Project Cybersyn, Chile, 1971–73

Recreation of the operations room, Museu del Disseny, Barcelona

AI and central planning

Jack Ma, speech to Zhejiang entrepreneurs, Shanghai, November 2016

"Over the past 100 years, we have come to believe that the market economy is the best system, but in my opinion, there will be a significant change in the next three decades, and the planned economy will become increasingly big. Why? Because with access to all kinds of data, we may be able to find the invisible hand of the market."



Jack Ma
(b. 1964)

AI and the calculation problem

- Computing with data is not the same as creating them. Bids, purchases and competing business experiments reveal which uses people value.
- Without markets for productive assets, an AI lacks the price and profit signals needed to compare competing plans.
- Firms and consumers can use AI too. A more powerful planner faces a changing economy, not a fixed spreadsheet.

Boettke and Candela (2023); Tabarrok (2018).

The Entrepreneur

What is an entrepreneur?

- It is harder to define than you might think. . .

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  - Is a Subway franchise owner an entrepreneur?

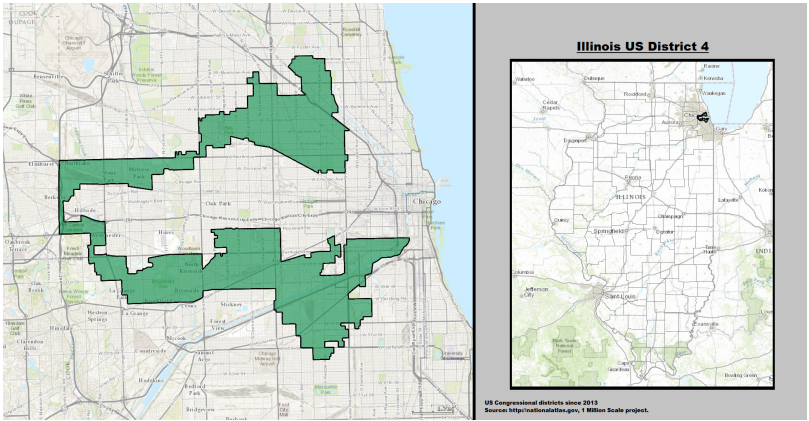
What is an entrepreneur?

- It is harder to define than you might think...
  - Is an inventor working inside an existing company an entrepreneur?
  - Is a Subway franchise owner an entrepreneur?
- What about non-market cases?

Military entrepreneurship?

Charitable entrepreneurship?

Political entrepreneurship



Illinois's 4th district, 2013–2023

Possible definitions

The traditional definition

An entrepreneur is someone who organizes, manages, and assumes the risk of a business or enterprise.

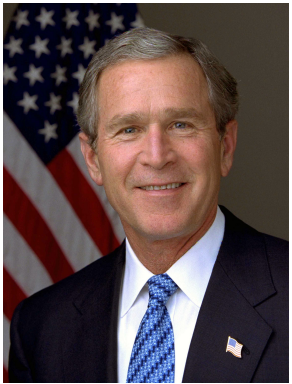
The broad definition

An entrepreneur is an agent of change.

The origin of the word

"The problem with the French is that they don't have a word for *entrepreneur*."

— George W. Bush, to Tony Blair?



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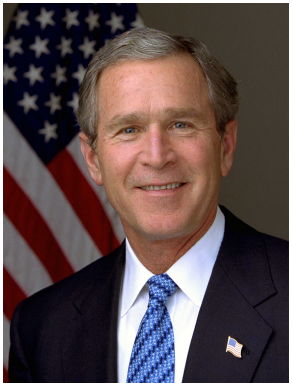


- The story comes from Shirley Williams, in 2002. Blair's office denied it. $\Rightarrow$ Probably not true.
- The word is French: *entreprendre*, to undertake.

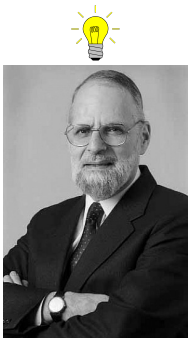
Cantillon, Say, and the entrepreneur

- Richard Cantillon (c. 1730): the entrepreneur buys at a certain price to sell at an uncertain one.
- Jean-Baptiste Say (1803): the entrepreneur makes the production decisions. He combines land, labor and capital into a product.
- He contracts with the capitalist (interest) and the workers (wages), and keeps what is left. $\Rightarrow$ Not necessarily a capitalist himself.

Jean-Baptiste Say
(1767–1832)

Kirzner and alertness

- Entrepreneurship is alertness to profit opportunities that others have not noticed.
- A price gap is a $10 bill lying on the sidewalk. The entrepreneur is the one who sees it.
- To have a lightbulb moment, you need to know what you are looking for. $\Rightarrow$ Profit tells you what to look for.

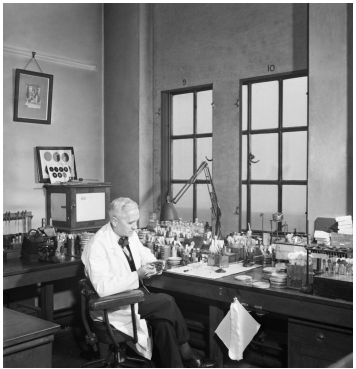


Israel Kirzner
(b. 1930)

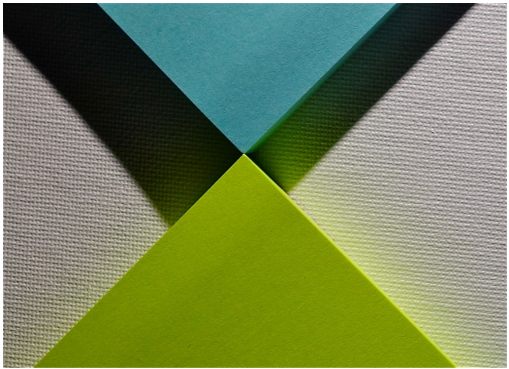
Bottoms Up draft beer

Bottoms Up: another profit opportunity

Serendipity and alertness



Penicillin, 1928: an accident in a lab



Post-it notes, 1980: a failed glue finds a use

Predicting the value of an innovation: the Internet

Paul Krugman, "Why Most Economists' Predictions Are Wrong," *Red Herring*, 1998

"By 2005 or so, [. . .] the Internet's impact on the economy has been no greater than the fax machine's."

Paul Krugman

Predicting the value of an innovation: broadcasting

E. E. Free, “Radio’s Real Uses,” *The Forum*, 1926

“Radio broadcasting is spectacular and amusing but virtually useless.”

Testing Hayek and Kirzner

- 135 students, 3 minutes each.
- Move water to an empty bucket 15 feet away, in a single trip.
- $12 for showing up, plus up to $1 for each cup moved.

Source: Demmert, Henry and Daniel B. Klein. (2003). “Experiment on Entrepreneurial Discovery: An Attempt to Demonstrate the Conjecture of Hayek and Kirzner,” *Journal of Economic Behavior & Organization* 50(3), 295–310.

Testing Hayek and Kirzner

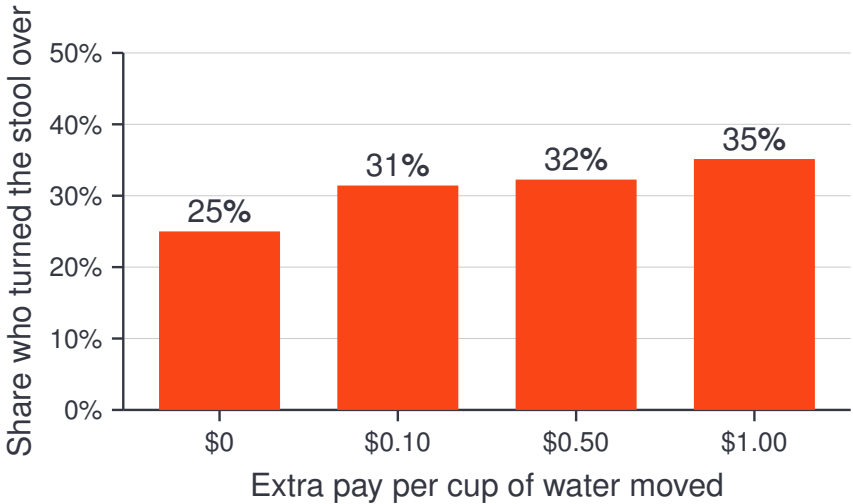
Photo 2. The obvious method of transferring water.

The obvious method

(b)

Photo 3. The nonobvious and much superior method of transferring water.

The superior method



Note: Share of the 135 subjects who turned the stool upside down, by payment schedule. Demmert and Klein (2003), Table 1. The differences are not statistically significant.

Schumpeter and creative destruction

Joseph Schumpeter, *Capitalism, Socialism and Democracy*, 1942

"The opening up of new markets, foreign or domestic, and the organizational development from the craft shop and factory to such concerns as U.S. Steel illustrate the same process of industrial mutation—if I may use that biological term—that incessantly revolutionizes the economic structure from within, incessantly destroying the old one, incessantly creating a new one. This process of Creative Destruction is the essential fact about capitalism."

Joseph Schumpeter
(1883–1950)

THE SATURDAY EVENING POST
THE SATURDAY EVENING POST
ALL FORD MODELS INTERCHANGEABLE
Every Model T FORD is an exact duplicate of the car that won the New York Seattle Contest
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Ford Motor Company
Detroit Manufacturers A.L.A.M.M.A.
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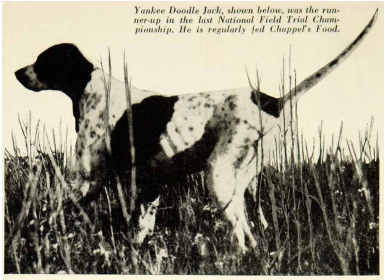
Yankee Doodle Jack, shown below, was the runner-up in the last National Field Trial Championship. He is regularly fed Chappel's Food.



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THE DOG FOOD SUPREME

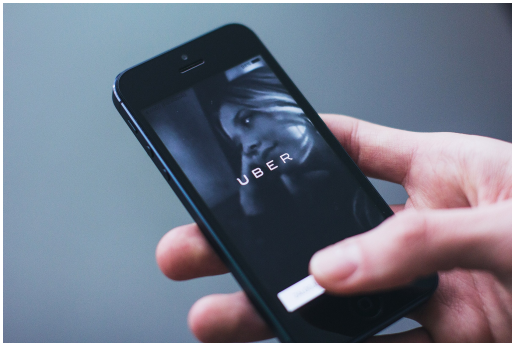
A&P vs. Walmart

A&P: the largest retailer in the world for decades

Walmart: founded 1962

Taxis vs. Uber

A New York taxi medallion: about $1 million in 2014

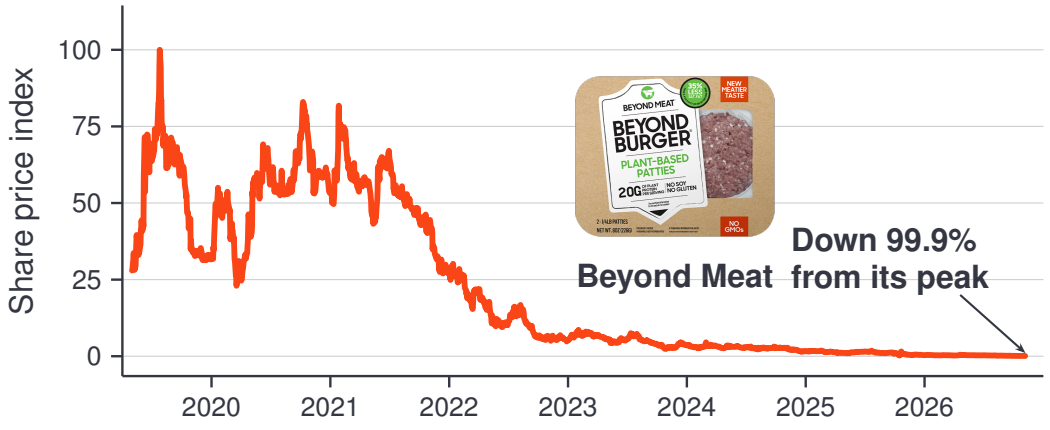


Uber in New York since 2011

Profit and loss

- Creative destruction is not good in itself. It is efficient only when it passes the profit-and-loss test.
- A profit says consumers value the output more than the inputs it used up. A loss says the opposite.
- Losses shut down the bad ideas. $\Rightarrow$ Subsidies and bailouts keep them alive, and the destruction comes without the creation.

An invention without a market



Note: Beyond Meat (BYND), daily closing share price, July 26, 2019 peak = 100. Split-adjusted prices through November 6, 2026; Yahoo Finance. Product photo: Beyond Meat.

Are financial markets shortsighted?

- A share is a claim on future profits. A company can be valuable even while it loses money today.
- Investors fund young firms because they expect future products and customers. High valuations can reflect patience, as well as optimism.
- Beyond Meat shows that these expectations can be wrong. A failed forecast does not, by itself, show that investors ignored the long run.

Why is the entrepreneur the residual claimant?

- Workers get wages and lenders get interest, fixed in advance. The entrepreneur gets whatever is left: the residual.

Why is the entrepreneur the residual claimant?

- Workers get wages and lenders get interest, fixed in advance. The entrepreneur gets whatever is left: the residual.
- Manne (2014): the entrepreneur's input is an idea. No one can measure what it is worth before it is tried, or monitor it while it is used.
- So it cannot be bought for a wage. The only way to be paid for it is to own the venture and keep the profit, or bear the loss.

Source: Manne, Henry G. (2014). "Resurrecting the Ghostly Entrepreneur," *Review of Austrian Economics* 27(3), 249–258.